

Advanced (Carolina Reaper)

- Q1.** Solve the system of inequalities $x + y \geq 2$ and $x - y \leq 1$. What is the intersection point of the boundary lines?
- (A) (1, 1)
(B) (0, 2)
(C) (2, 0)
(D) (3, -1)
(E) (-1, 3)
- Q2.** A bakery sells a total of 120 loaves of bread per day. They sell a combination of whole wheat and white bread. If x represents the number of whole wheat loaves and y represents the number of white bread loaves, which inequality represents the total number of loaves sold?
- (A) $x + y \leq 120$
(B) $x + y \geq 120$
(C) $x + y = 120$
(D) $x - y \geq 120$
(E) $y - x \leq 120$
- Q3.** If x and y satisfy the system of inequalities $x + y \geq 4$ and $x - y \geq 2$, what is the minimum value of x ?
- (A) 3
(B) 4
(C) 5
(D) 6
(E) 7
- Q4.** A company produces two products: A and B . The production of A requires 2 hours of labor and 3 hours of machine time, while the production of B requires 3 hours of labor and 2 hours of machine time. If the company has 240 hours of labor and 210 hours of machine time available per week, which system of inequalities represents the production constraints?
- (A) $2x + 3y \leq 240$ and $3x + 2y \leq 210$
(B) $2x + 3y \geq 240$ and $3x + 2y \geq 210$
(C) $2x + 3y = 240$ and $3x + 2y = 210$
(D) $2x - 3y \leq 240$ and $3x - 2y \leq 210$
(E) $2x - 3y \geq 240$ and $3x - 2y \geq 210$
- Q5.** Solve the inequality $3x - 2y > 5$. What is the equation of the boundary line?
- (A) $3x - 2y = 5$
(B) $3x + 2y = 5$
(C) $2x - 3y = 5$
(D) $2x + 3y = 5$
(E) $x - y = 5$
- Q6.** A car rental company has 50 cars available for rent. The company charges 40 per day for a compact car and 60 per day for a full-size car. If x represents the number of compact cars rented and y represents the number of full-size cars rented, which inequality represents the total number of cars rented?
- (A) $x + y \leq 50$
(B) $x + y \geq 50$
(C) $x + y = 50$
(D) $x - y \geq 50$
(E) $y - x \leq 50$
- Q7.** A farmer has 100 acres of land available for planting. The farmer wants to plant a combination of wheat and oats. If x represents the number of acres of wheat and y represents the number of acres of oats, which system of inequalities represents the total amount of land available?
- (A) $x + y \leq 100$ and $x \geq 0$
(B) $x + y \geq 100$ and $y \geq 0$
(C) $x + y = 100$ and $x \leq 0$
(D) $x - y \leq 100$ and $y \leq 0$
(E) $y - x \leq 100$ and $x \geq 0$
- Q8.** A company produces two products: A and B . The production of A requires 2 units of material X and 3 units of material Y , while the production of B requires 3 units of material X and 2 units of material Y . If the company has 240 units of material X and 210 units of material Y available, which inequality represents the total amount of material X used?
- (A) $2x + 3y \leq 240$
(B) $2x + 3y \geq 240$
(C) $2x + 3y = 240$
(D) $2x - 3y \leq 240$
(E) $2x - 3y \geq 240$

Open Ended Questions (FRQ)

- Q9.** A company produces two products: X and Y . The production of X requires 3 hours of labor and 2 hours of machine time, while the production of Y requires 2 hours of labor and 3 hours of machine time. If the company has 240 hours of labor and 210 hours of machine time available per week, determine the maximum number of units of X and Y that can be produced. Show all work and explain your reasoning.
- Q10.** A city has a budget of 100,000 for a new park. The park will have a combination of playground equipment and walking trails. If the playground equipment costs 10,000 per unit and the walking trails cost 5,000 per unit, determine the maximum number of units of playground equipment and walking trails that can be purchased. Show all work and explain your reasoning.

Chef's Tips

- Tip 1: Make sure to read each question carefully and understand what is being asked.
- Tip 2: Use the given information to determine the constraints and the objective function.
- Tip 3: Show all work and explain your reasoning for each problem.